Reza Mahjourian

Website, Google Scholar

RESEARCH PROFILE

❖ Research scientist specializing in foundation models for robotics and autonomous systems, with a track record of translating frontier research into deployed AI systems.
❖ Expertise in large-scale world modeling, multi-agent prediction, and applying large-scale vision and vision-language models to robotics, autonomous driving, and safety-critical systems.
❖ Author of 13 patents and publications in CVPR, ECCV, ICRA, IROS, and AAAI with 2500+ citations.
❖ Selected contributions include saliency-guided foundation models, occupancy flow fields for motion prediction, scalable scenario generation for autonomous systems, sample-efficient reinforcement learning for robot control, and self-supervised learning of depth and ego-motion.

SKILLS

A.I. RESEARCH

- Foundation Models
- Robotics
- Computer Vision
- Multi-Agent Prediction
- Reinforcement Learning
- Evolutionary Strategies

EDUCATION

UNIV. OF TEXAS AT AUSTIN

PhD in Computer Science
GPA: 4.0

SHARIF UNIV. OF TECH.

BS in Computer Eng.
Tehran, Iran

OPEN SOURCE

❖ **GameGraph**: Studies impact of domain stochasticity and ergodicity on reinforcement learning.
❖ **Discovery**: Evolutionary feature discovery for reinforcement learning.
❖ **OpenNERO**: Platform for A.I. research and education.

EXPERIENCE

WAYMO RESEARCH

Jul 2023 - Feb 2026 | Staff Research Scientist
Jul 2019 - Jul 2023 | Senior Research Scientist

- Led development of a saliency-guided vision preprocessing framework enabling efficient fine-tuning of Gemini multimodal foundation models for safety-critical autonomous driving tasks such as detection and segmentation, improving long-tail object detection by 5–34% and segmentation accuracy by 10–16%.
- Led research on scalable occupancy and trajectory prediction. Established occupancy modeling as a shared representation across prediction, simulation, and planner evaluation pipelines, and led cross-functional efforts with Perception and Simulation teams to deliver two production systems: the onboard Occlusion Prediction System and a Scenario Generation pipeline for synthesizing novel driving scenarios.
- Founded and organized the Waymo Open Dataset Occupancy & Flow Prediction and Scenario Generation challenges at CVPR (**2022**, **2024**, **2025**), engaging 20+ academic and industry teams and catalyzing 15+ follow-up peer-reviewed publications across academia and industry.

GOOGLE BRAIN ROBOTICS | Student Researcher

Sep 2017 - Nov 2018 | Mountain View, CA

- Developed sample-efficient reinforcement learning methods for robot table tennis using self-play (**ArXiv 2018**, **Website**.) The hierarchical and modular policy design was later adopted in a multi-year project achieving human-level competitive robot table tennis (**RSS 2023**, **ICRA 2025**, **Videos**).
- Unsupervised learning of object depth and motion from raw monocular videos. **AAAI 2019**, **CVPR 2019**, patent, **Website**, **Google AI Blog**.
- Future semantic segmentation using 3D structure, **ECCV 2018**, patent.
- Added computer vision library functions to **TensorFlow**.

GOOGLE BRAIN ROBOTICS | Research Intern

May 2017 - Aug 2017 | Mountain View, CA

- Developed a novel self-supervised learning method for depth and ego-motion from monocular videos. **CVPR 2018**, **Website**, patent, patent, patent.
- Developed custom TensorFlow op for aligning 3D point clouds during training.

GOOGLE BRAIN | Research Intern

May 2016 - Aug 2016 | Mountain View, CA

- Geometry-based next frame prediction from monocular video.
- IEEE Intelligent Vehicles Symposium 2017 **Paper**, patent, patent.

GOOGLE | Software Engineering Intern

Jun 2015 - Aug 2015 | Mountain View, CA

- Designed large-scale neural models for predicting in-app purchases using full Google Play historical data. Achieved measurable lift over production baselines.

TEAMUP | Start-up Lead Developer

Jul 2009 - Dec 2009 | London, UK

- Created a Django web app for managing fitness and sports businesses.
- Led the project from inception to beta release **Website**